APPLIED MACHINE LEARNING FINAL PROJECT

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Part 1a – Approach

- Model Section (Traditional): Gradient Boosting + PCA
  - StratifiedKFold CV (k=5) used to test multiple models
    - Logistic Regression, KNN, SVM...
    - Performance metric was accuracy over all 5 folds
  - PCA to reduce dimensionality
    - 48 dimensions to 31 dimensions
  - Standardscalar to normalize data
  - Used GirdSearchCV for some models

Part 1a – Results

- Five-fold CV results (80%)
  - Mean accuracy of 0.759
  - Mean F1 score of 0.758
- 20% held out set
  - Mean accuracy and F1 score of 0.760
- Predicted Results on test set
  - Similar to held out set

Part 1b – Approach

- Model selection: Multi-Layer Perception (MLP)
  - Two hidden layers, 128 and 64 nodes
  - BatchNorm after each layer
  - Activation function was ReLU
  - Dropout (0.3)
  - Adam used for learning rate
  - Performance validated with StratifiedKFold (k=5)

Part 1b – Results

- Five-Fold CV results (80%)
  - Accuracy: 0.761
  - F1: 0.759
- Held out 20%
  - Accuracy: 0.771
  - F1: 0.769
- Predicted Results:
  - Similar to held out set.

Part 1 – Conclusion

- Training time is costly
- PCA is powerful
- Worth tuning hyperparameters?

Part 2 – Approach

- Model Selection (Traditional): LogReg
  - StratifiedKFold CV used to determine best model
  - StandardScalar used to normalize data
  - SelectKBest used to reduce dimensionality

- Model Selection (Deep Learning): MLP
  - RepeatedStratifiedKFold CV used to determine hyperparameters
  - StandardScalar used to normalize data
  - SelectKBest used to reduce dimensionality
  - ReLU with 0.25 dropout
  - BatchNorm used to normalize activations
  - BCEWithLogitsLoss used to measure error

Part 2 – Results

- Deep learning model selected for analysis
- Fifteen-fold repeated CV results (80%)
  - Mean AUC-ROC of 0.706
  - Mean F1 score of 0.758
- Performance on training set (predicted Results on test set)
  - Mean accuracy of 0.7658
  - Mean F1 score of 0.6823

Part 2 – Conclusion

- High dimensional, low sample count data is difficult to work with
  - Synthesizing data may not always work
- Deep learning can be highly adaptable in ways even the best traditional learning methods aren't

THANK YOU